

Ankit Kumar Ojha

AI Engineer — Python — FastAPI — LangChain — LangGraph — LangSmith — MCP

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github : <https://github.com/ankitojha15>

Professional Summary

AI Engineer (B.Tech Data Science, 2025) with hands-on experience building and shipping RAG systems and AI agents using Python. Skilled in LangChain, LangGraph, LangSmith, FastAPI, SQL/Postgres, FAISS, Docker, MCP and Tool Calling. Built live projects including DriftGuard AI (28/30 auto-fixed, re-runs down 60%), Veritas AI with page-level citations and evals (Hit@5, MRR, NDCG), and VideoRAG for YouTube Q&A. Looking to build practical and reliable AI solutions in production.

Projects

DriftGuard AI — Self-Healing Data Pipeline — Live Sep 2026

GitHub: <https://github.com/ankitojha15/DriftGaurd-AI> | Live: <https://driftgaurd-ai-night-guard.streamlit.app/>

- Tech Stack: LangGraph, LangChain, Groq, Postgres (Neon), FastAPI, Streamlit, Slack, Render.
- Detect schema drift by diffing Postgres information_schema between prod and sandbox copies.
- Draft fix SQL with Groq (LangChain), test on sandbox with rollback on fail.
- Apply to prod only after human approval in Streamlit (FastAPI /detect, /approve).
- Live with daily cron and Slack alerts; **28/30 golden breaks auto-fixed, re-runs down 60%**.

Veritas AI — Hallucination-Guarded RAG over PDFs — Live Sep 2026

GitHub: <https://github.com/ankitojha15/Veritas-AI> | Live: <https://knowledgebaseai-pzi7.streamlit.app/>

- Tech Stack: Python, LangChain, FAISS, Hugging Face Embeddings, Groq, FastAPI, Streamlit.
- Built LangChain RAG system answering PDF questions with page-level citations using Hugging Face embeddings, parent-child chunking, hybrid dense + BM25 retrieval with RRF and cross-encoder reranking.
- Used Groq with query rewriting and citation verification; built FastAPI + Streamlit chat UI with streaming.
- Built 60-query eval harness (50 normal + 10 adversarial) reporting **Hit@5, MRR, NDCG**; tested end-to-end.

VideoRAG — YouTube Video Q&A System — Live Aug 2026

GitHub: <https://github.com/ankitojha15/VideoRAG> | Live: <https://videorag-7swz.onrender.com>

- Tech Stack: Python, FastAPI, FAISS, BGE ONNX, Groq, Gemini, Chrome Extension, Render.
- Built system for answering questions from YouTube transcripts; Chrome extension + FastAPI backend on Render.
- Used BGE ONNX embeddings, FAISS (MMR), Groq LLM for retrieval and answering with Hindi-to-English translation via Gemini.

X Post Generator Aug 2026

GitHub: <https://github.com/ankitojha15/LangGraph>

- Tech Stack: LangGraph, Python, LLMs.
- Built AI-powered post generation agent using LangGraph and LLMs that iteratively generates, evaluates, and refines posts based on defined quality criteria.
- Implemented stateful workflows, conditional routing, iteration control, and human-readable output for automated content generation.

Technical Skills

- **Languages:** Python, SQL
- **Data/ML:** Scikit-learn, NumPy, Pandas
- **AI/GenAI:** LangChain, LangGraph, LangSmith, RAG, AI Agents, MCP, Tool Calling, Memory, Observability, Prompt Engineering, LLM Evals
- **Backend:** FastAPI (REST, Async), SQLAlchemy, Pydantic
- **Data/AI Infra:** FAISS, Qdrant, Pinecone, BGE / HuggingFace Embeddings, Hybrid Retrieval (BM25, RRF, Reranking), Groq, OpenAI, Gemini APIs, Postgres (Neon, pgvector), MySQL
- **LLM Fundamentals / Inference:** Tokenization (BPE), Attention (MHA, GQA, FlashAttention), KV Cache, Prefill / Decode, PagedAttention, vLLM, Quantization, Streaming
- **Tools:** Git, GitHub, Docker, AWS, Render

Education

Oriental Institute of Science & Technology, Bhopal
B.Tech - Data Science

Sep 2021 – Jun 2025